1 A disc moves on a smooth horizontal surface. Fig. 1.1 shows the top view of the disc moving from point P at $t = 0.0$ s with an initial velocity of 0.200 m s^{-1} in the north-east direction. From $t = 0.0$ s to $t = 10.0$ s, the disc has a constant acceleration of 0.100 m s^{-2} in the easterly direction.

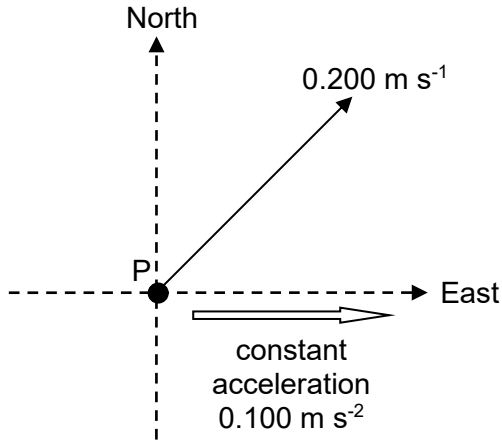


Fig. 1.1

(a) Determine the disc's velocity at t = 10.0 s.

speed = ……. m s^{-1}

bearing = …… ° from north [3]

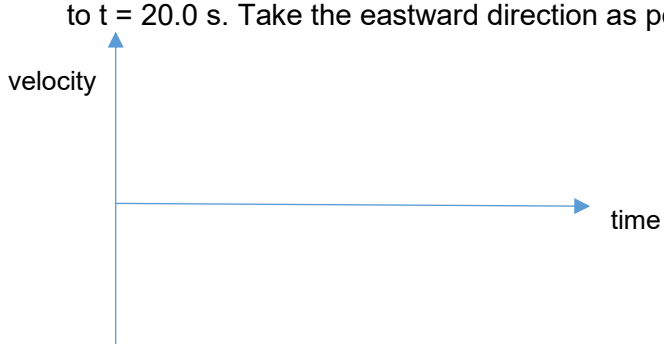
[Turn over

(b) Determine the distance of the disc from the point P at t = 10.0 s.

distance = …….m [3]

(c) From $t = 10.0$ s to $t = 20.0$ s, the acceleration of the disc has a constant value of 0.100 m s^{-2} directed towards the west.

1. Sketch the velocity-time graph along the East-West direction between t = 10.0 to t = 20.0 s. Take the eastward direction as positive.



[2]

2. Sketch in Fig. 1.2 the path of the disc from t = 0.0 s to t = 20.0 s.

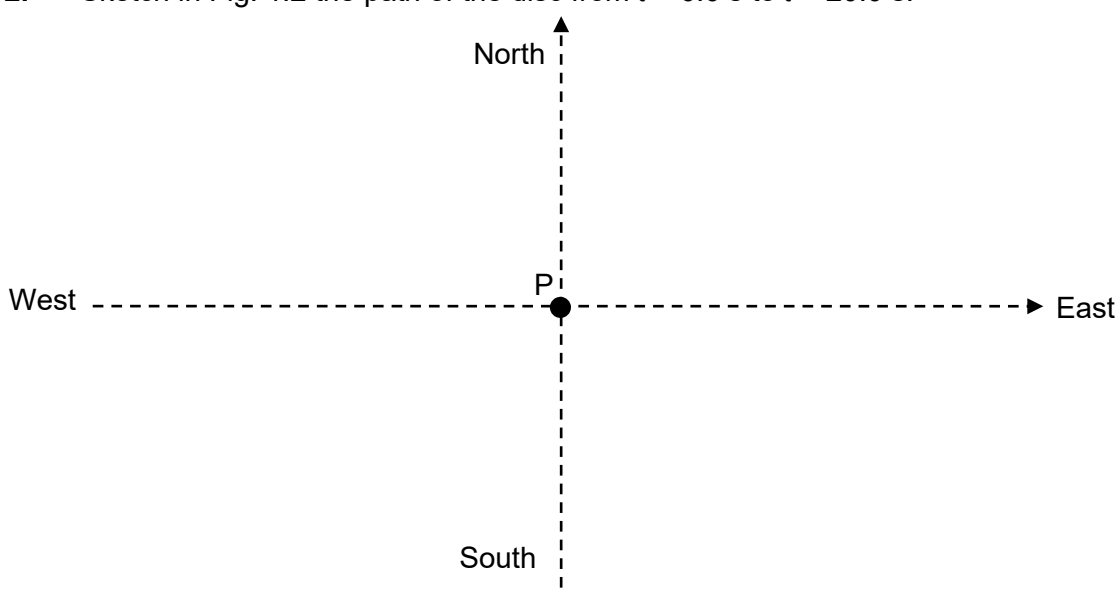


Fig. 1.2

[2]

[Turn over